Polynomial addition

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#include <stdio.h>
#include <stdlib.h>

struct Node {
   int coeff, exp;
   struct Node *next;
};

struct Node* insert(struct Node *head, int c, int e) {
   struct Node *p = malloc(sizeof(struct Node));
   p->coeff = c;
   p->exp = e;
   p->next = NULL;

   if (head == NULL)
      return p;

   struct Node *t = head;
   while (t->next != NULL)
      t = t->next;

   t->next = p;
   return head;
```

```
}

struct Node* add(struct Node *p, struct Node *q) {
   struct Node *r = NULL;

   while (p && q) {
      if (p->exp == q->exp) {
         int sum = p->coeff + q->coeff;
         if (sum != 0)
            r = insert(r, sum, p->exp);
         p = p->next;
         q = q->next;
      }
      else if (p->exp > q->exp) {
         r = insert(r, p->coeff, p->exp);
         p = p->next;
      }
      else {
         r = insert(r, q->coeff, q->exp);
         q = q->next;
      }
   }

   while (p) {
      r = insert(r, p->coeff, p->exp);
      p = p->next;
```

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    }

    while (q) {
        r = insert(r, q->coeff, q->exp);
        q = q->next;
    }

    return r;
}

void display(struct Node *p) {
    while (p) {
        printf("%dx^%d", p->coeff, p->exp);
        p = p->next;
        if (p) printf(" + ");
    }
    printf("\n");
}

int main() {
    struct Node *p = NULL, *q = NULL, *r;

    p = insert(p, 5, 4);
    p = insert(p, 3, 2);
    p = insert(p, 7, 0);
```

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    q = insert(q, 2, 4);
    q = insert(q, 6, 3);
    q = insert(q, 3, 2);
    q = insert(q, 1, 0);

    r = add(p, q);

    printf("P1 + P2 = ");
    display(r);

    return 0;
}
```

Output :P1 + P2 =

7x^4 + 6x^3 + 6x^2 + 8x^0